CHIGARAPALLI SHRUTHI REDDY

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WORK EXPERIENCE:

Telecommunication Network Engineer: **March 2016 - Present**

Quadgen Wireless Solutions Inc., King of Prussia, PA

- Performing integrations in 3GPP RAN/eRAN which includes assigning SAID IP address on Cisco/ALU routers for any Radio Node or New Site Builds
- Responsible for providing XML and MoShell scripts to define radios and other ericsson network elements required for integrating 4G LTE/UMTS carrier on a new/existing EnodeB
- Thorough knowledge on Packet Core Networks to troubleshoot nodes and align Radios and EnodeB related parameters as per AT&T's standards
- Support Field Engineers & Turf vendors for integration and VoLTE Call Tests

Graduate Research Student **January 2016**

Toshiba Stroke & Vascular Research Centre, State University of New York, Buffalo, NY

- Calculated the severity of the stenosis my analyzing Fractional Flow Reserve (FFR) using digital pressure sensors across the stenosis in a 3D printed vascular phantom model
- The results were acquired using piezo-resistive pressure on the phantoms and the data was formulated using LabVIEW and a PCB board

Internship

Sify Corp, Chennai, India **June – August 2015**

- Department of Postal Office (DOP): A redundant structure for Data Centre was designed for the routers to follow MPLS and we implement the MPLS firewall and MPLS IPS. Routing Protocols, BGP and OSPF were used for this system
- Voice over IP: Gained knowledge on SS7 Network & the entire structure of VoIP. Worked on SIP
- Wireless Engineering Testing Team: Tested the Last Mile by configuring the VLAN modes and setting up virtual routers to check the optimized data rate by pumping the full capacity traffic through

TECHNICAL SKILLS:

Programming Skills:	C, Python, Data Structures, XML & MOShell Scripting
Simulators:	Wireshark, NS-3, Eclipse, Quartus, MATLAB, Simulink
Protocols and Standards:	TCP/IP, UDP, AdHoc, Ethernet, BGP, OSPF, DHCP, EIGRP, VPN, OLSR, GPRS, OFDM, DNS, Bluetooth, 802.11 a/b/g/n/ac, MPLS
Wireless Architecture:	LTE, IMS, FMC, UMTS
Network Platform:	PuTTy, Ericsson OSS (GUI & CLI), AOTS, MBBTS Noise tool, Ni Gaurdian

EDUCATION:

Master's in Electrical Engineering **February 2016**

University at Buffalo, The State University of New York

GPA: 3.7/4

Bachelor's in Electronics & Communication Engineering **June 2014**

Amrita Vishwa Vidyapeetham, Bangalore, India

GPA: 8.59/10

RELEVANT COURSEWORK:

Multi Input Multi Output Communications, Principles of Networking, Embedded and Network System Design, Cellular and Mobile Communications, Digital Signal Processing

TECHNICAL EXPERIENCE:

Implementation of UDP using DE2i-150 boards **December 2015**

State University of New York, Buffalo, NY

- Implemented NIOS II Hardware system with Triple- Speed Ethernet IP Core on Altera's DE2i-150 FPGA board using Qsys and synchronized Marvell 88E111 Ethernet PHY chip and other components of the FPGA board
- Walkie Talkie application using UDP Protocol was developed in Eclipse using DE2i-150 FPGA board which was programmed to behave as transceiver
- Application performed actions such as force exit if pre-defined emergency phrase is detected

Lossless Encoding of an I-frame **Apr-May 2015**

State University of New York, Buffalo, NY

- Encoded an I-frame using JPEG Technique to reduce the compression errors. An 8X8 DCT (Discrete Cosine Transform) is applied on the frame so as to achieve high energy compaction and then Zig-Zag Reordering of these coefficients sort the low frequency and high frequency in the order of the signal energies
- Zero Length Coding is applied on AC values and Difference Coding method is applied on DC value. We perform Huffman Coding (w.r.t. standard Huffman Table) on these coefficients. This results in the encoded frame

Baby Monitoring System **January-May 2014**

Amrita School of Engineering, Bangalore, Karnataka, India

- Program was developed using MATLAB GUI and Terminal as interface and designed to assimilate baby cries such as Hunger, Tiredness and Agitation and notify the cause based on coefficients of various feature analysis algorithms like Magnitude Sum Function, Zero Crossing and Pitch Detection
- Published a paper in the International Journal of Scientific and Technology Research (IJSTR)

(Reference No.: IJSTR-0614-9345)